

Machine Learning Techniques for Credit Scoring in Banking with Management, HR, and Organizational Key Components

Dr.Niti Saxena¹, Dr. S. Thandayuthapani², Dr. P. Raman³, Dr.ThirumaValavan A⁴, Dr.Nihar Ranjan Behera⁵ and Dr.Eti Khatri⁶

¹Associate Professor, Management & Commerce, Jagannath International Management School, New Delhi, India,

²Assistant Professor, Vel Tech Rangarajan Dr.Sagunthala, R&D Institute of Science and Technology, Tamilnadu, India,

³Department of Management Studies, Panimalar Engineering College, Chennai,

⁴Deputy Director, Indian Institute of Banking & Finance, Mumbai, Maharashtra, India,

⁵Adjunct Faculty, SSBM Geneva, Swiss School of Business and Management, , GBC - Geneva Business Center, Genève, Switzerland

⁶Associate Professor, Department of Management Studies, Nitte Meenakshi Institute of Technology, Bengaluru, Karnataka, India

E-mail : niti.saxena@yahoo.com, sivakulanthay@gmail.com, raams.ram70@gmail.com, thirumavalavan63744@gmail.com, nihar1773@gmail.com, etikhatr2007@gmail.com

Abstract- Credit rating is crucial in the fast-changing 21st-century banking industry to determine creditworthiness. Traditional credit score systems may not be able to handle today's complex money habits because they are focused on statistics and prior data. This research advises adding management, human resources, and organizational factors to machine learning credit evaluations in addition to financial data. Structure of the research describes different machine learning types. Logistic regression, decision trees, random forests, gradient boosting, and neural networks. The algorithms are trained using this dataset's financial metrics, management practices, HR indicators, and organizational procedures. Feature engineering strategies pull data from various sources to get a full picture of someone's reputation. The research argues that machine learning models should be transparent, especially in the highly regulated banking business. Using LIME and SHAP values helps make credit scoring determinations more dependable and understandable. Credit scoring will be more precise, and financial institutions will understand credit risk aspects better. Banks can improve loan selections, portfolio performance, and risk by adding management, human resources, and organizational data to financial data. This research helps financial organizations analyze credit risk in the age of machine learning and big data, resulting in more accurate credit score models.

Keywords: Credit Scoring, Machine Learning, Banking Management, HR (Human Resources), Organizational Key Components, Risk Assessment

I. INTRODUCTION

Credit scores are extremely significant in the fast-paced world of banking and finance since they demonstrate the borrower's level of dependability from a financial perspective. Methods that are based on rules and mathematics have been utilised in the past for the purpose of credit evaluation. On the other hand, the emergence of technologies that have machine learning capabilities has

provided financial institutions with more advanced methods to evaluate credit risk [1].

The purpose of this research is to investigate the application of machine learning in the banking sector for the purpose of credit evaluation. More specifically, the research focuses on the incorporation of management, human resources, and other crucial aspects of the business into the models. It is possible that banks will improve the accuracy and reliability of their credit scoring algorithms in order to improve their lending decisions. Information pertaining to organisational structures, human resource standards, and management practices is utilised by these management systems.

When conducting a credit assessment, it is important to take into consideration management, human resources, and organisational characteristics for a variety of reasons. Important information regarding the health and success of the borrower's business can be gleaned from these commodities. Those who lend money are more inclined to provide loans to companies that have a solid organisational structure and good management than they are to make loans to companies that have poor management and an unstable organisational structure. Through careful examination of these aspects, financial institutions are able to ascertain the state of the borrower's financial situation and identify the elements that influence the borrower's capacity to repay the loan.

In this research, we will investigate the issues that are associated with the traditional methods of credit rating, with a particular emphasis on the issues that machine learning techniques attempt to resolve. Next, we will discuss the advantages and disadvantages of the most prominent machine learning approaches that are utilised in the process of credit scoring. In addition to this, it will

be meticulously gathered, processed, and incorporated into data that is essential for the administration, staff, and structure of the credit rating system.

The purpose of this research is to investigate the potential advantages that could be gained by incorporating machine learning and organisational data into credit scoring systems that are utilised in the financial sector. The utilisation of contemporary analytics and a profound comprehension of the functioning of companies can assist financial institutions in enhancing their risk assessment processes, reducing the likelihood of default, and fostering more sustainable lending practices.

II. RELATED WORK

Looking at credit scores and risk using machine learning methods [2].

Nonperforming loans are becoming more common very quickly in all parts of Kenya's financial system, even in microfinance organizations. In its report for the first quarter of 2023, the Central Bank of Kenya says that credit risk is the biggest concern that affects the security of Kenya's financial institutions and the system as a whole. The Central Bank of Kenya (CBK) knows that credit risk is a big problem because banks' main job is to give money. To keep up with the growing number of loans, the banking industry needs to quickly adopt and improve technologies for credit scoring and risk assessment [3].

In the financial industry, machine learning methods are reliable and new ways to figure out credit risk, spot fraud, and make credit scores. Machine learning systems that look at past loan data, client behaviour, credit scores, income, and other relevant factors can figure out how likely it is that a loan will not be paid back. Microfinance and the banking sector have a hard time making accurate models because they don't have enough data. Because of this, it is very important for each bank's credit research teams to work closely with the outside companies they've hired to make credit scoring models [4]. When this strategy is used correctly, it provides a complete system that lowers the risk of loan default by using datasets that cover a lot of different areas.

Another big problem for the banking industry is figuring out how to deal with borrowers' changing financial situations. By looking at data from the past and the present, machine learning systems might be able to guess what trends will happen in the future. This makes it possible to create early warning systems that can spot signs of financial trouble or changes in the borrower's finances. One way for a bank to lower the risk of failure is to quickly spot signs of trouble and take action, like renegotiating loan terms.

Techniques like unsupervised learning have also been shown to work well in the retail business for customizing products and dividing customers into groups. Similar methods may be used in the financial sector to put customers into groups based on their risk levels, spending habits, and general financial behaviour. Decisions about proposals, loan limits, and interest rates can be changed based on the risk of default in a certain area [5].

This piece looks at how machine learning can be used to solve problems in the banking industry. Is your business having trouble with credit scoring systems that don't work right and loans that aren't paid back? Our opinion is that the future of ethical lending will depend a lot on the use of new ways to check credit. We'd love to help you make the ways and processes you use to decide if your clients are creditworthy better.

III. RESEARCH METHODOLOGY

Credit scores are numbers that financial institutions use to quickly figure out if a business or person is creditworthy. The debtor's current and past payment history, the amount of money they still owe, how often they are late or miss payments, and other information from their credit report are all used to figure out the number. One way to tell if a company is creditworthy is to look at its financial situation [6]. Aside from business risk, many other things are looked at, such as operating efficiency, profitability, liquidity, solvency, and financial flows.

The lender is less likely to lose money if the credit score is good.

A Quick Look at Machine Learning To sum up, a review A branch of artificial intelligence called machine learning looks at data and figures out what it means. This is a piece of software that makes it easier to make analysis models. At its most basic, it is a system that can find patterns in data, learn from them, and make choices with little help from people. Machine learning techniques use computer methods to get information from data. There are no set formulas that these systems use to work; they work on their own.

In ML-driven scoring systems, the borrower's expected income potential, current earnings, and job chances are given more weight than their past performance, which is what traditional scoring systems use to decide if the borrower is creditworthy. That means that people who don't have much or any credit history can still get the cash they need by putting up collateral that shows they can pay their bills. Lending companies can now use the results of AI models to make more accurate profit projections.

When looking more closely, there are a lot of parallels with the idea of business grading.

Here's an example of how Machine intelligence (ML) could change the way that company credit scores are shown visually:

Smart data processing powered by ML Machine Learning is naturally good at quickly looking through large and various datasets. These databases have both structured and unstructured data in them, like news stories, social media posts, business trends, and information about the economy as a whole. One example of this is knowing what's going on in the market and what the latest news is about a company. These things can affect a company's creditworthiness but aren't shown in standard financial statements [7].

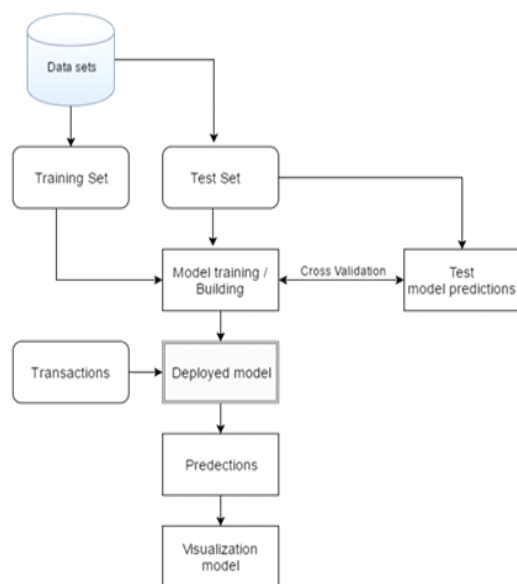


Fig.1 Denotes flowchart for credit scoring using machine learning.

In real time, a look at business credit worries Business credit scores have usually been based on cyclical financial data, which is why they aren't always changing. The addition of artificial intelligence has made it easier to send up-to-date information, such as the latest financial data, market conditions, and even changes to how things are done. ML can quickly change a company's credit rating to reflect changes like getting a big deal or losing money unexpectedly [8].

Predictive modeling is used by credit scoring companies. A form of artificial intelligence called machine learning makes models that can predict what will happen. These models look at past data and trends to guess how well a company will do financially in the future. With this skill, you can guess how credit risk and failure rates will change in the future. Treasurers can make better decisions with the help of ML because it can correctly predict things like a drop in business and a rise in credit risk.

Risk estimates that are tailored to each personal Machine Learning (ML) makes risk evaluation more personalized by taking into account the unique conditions, industry, and activity patterns of each company. For instance, AI can change the way credit is given to take into account the different ways that different types of companies, like tech companies and big industrial giants, show risk [9].

Implementing ML to Make Things Run More Smoothly It is faster to make decisions when artificial intelligence is used in the credit rating process. Credit report writers who are humans can take weeks to make a report, but AI systems can do the same thing in minutes or hours. This not only makes the Treasury Department work better, but it also gives business treasurers more control over their credit risk.

Making it easier to use artificial intelligence in risk management. Artificial intelligence (AI) can help corporate treasurers manage risks for their companies by giving them up-to-date credit scores, predicting how money will behave, and finding trends. When artificial intelligence predicts a failure, for example, treasurers can change their credit portfolios ahead of time to avoid high-risk exposures.

It is very important to protect data privacy and security because AI models need a lot of data to work properly. Collecting organizations must carefully follow data protection laws and make sure that sensitive information is handled in a private way. A breach can cause a lot of damage to your finances and image.

Deep learning-based ML models can work as "black boxes," meaning that no one knows how they work on the inside. It is very important to check how clear and easy to understand the model is. Credit rating choices should be open to everyone, including regulators and stakeholders. This might not always be possible with AI algorithms, though [10].

The quality and clarity of the data used to teach artificial intelligence have a big effect on how well it works. If you use wrong or biased data to get credit scores that aren't accurate, you might make bad financial decisions.

Making sure you follow the rules set by the government: Credit ratings are based on strict rules that don't always take into account how AI methods might affect things. So, businesses have to figure out a complicated set of rules when they try to use AI in credit scores.

Putting too much faith in Machine Learning (ML) and AI models: One bad thing that could happen if risk managers rely too much on ML programmes is that they might become too comfortable. AI has the potential to give us useful information, but we need to keep an eye on it, make sure that its results are correct through independent

analysis, and make sure that it uses good judgement [11]-[12].

Cost and Execution Dealing with Obstacles Adding artificial intelligence to a credit scoring system that is already in place could be hard and cost a lot of money. In addition to the cost of building and maintaining the AI system, there is also the likely need to train workers and make improvements all the time to keep up with the organization's changing needs.

IV. RESULTS AND DISCUSSION

For a long time, the evaluation of risk associated with biological or agricultural assets was mostly dependent on analytical hierarchy approaches and fuzzy logic. Both of these procedures are widely acknowledged to be fundamentally subjective in the field of research. On the other hand, due to the fact that it is a binary categorization method, machine learning continues to be the most effective method for evaluating the risk associated with agricultural assets. One of the challenges that farmers encounter when trying to acquire traditional loans is that they do not have sufficient collateral. Researchers looked at 1249 different samples of production and operation data in order to arrive at an assessment of the credit risk posed by new agricultural businesses in three different regions in China. An XG Boost model and many other machine learning methods, including logistic regression, support vector machines, and random forest, were subjected to a comparison analysis in order to determine which one was superior. The machine learning technique known as XG Boost, which is based on the gradient boosting decision tree (GBDT), is useful for effectively addressing the binary problem of determining whether or not the client would default on their payments. Educational attainment and the ratio of assets to liabilities are two crucial measures of credit risk for agricultural firms that are just starting out.

There are four common hazards that are associated with lending money to people in rural areas. These risks are human risk, operational risk, farm asset risk, and policy and market risk relating to agricultural products. Artificial intelligence (AI) and big data will play an increasingly important role in the decision-making process for loans. This is because AI and big data will help minimize risks and enhance the accuracy of default prediction models.

Using frameworks that are based on theories such as the diffusion of innovation theory, technology-organization-environment theory, and actor-network theory, a number of financial institutions are integrating their traditional and digital credit scoring techniques.

Financial institutions use machine learning (ML) to improve the objectivity of their lending choices in order to better serve their customers. Moreover, there is a view

that machine learning (ML) generates more sophisticated credit ratings and risk evaluations in comparison to the well-established credit scoring system of Fair Isaac Corporation (FICO). This belief is based on the fact that ML is able to offer more accurate results.

When it comes to credit rating, one of the challenges that arises is accurately distinguishing between debtors who are in default on their payments and those who are not in default. Researchers presented a modified harmony search random component in order to improve performance, explainability, and computing speed. This was done due to the hyper parameter sensitivity of black-box models such as SVM and RF.

Nevertheless, it is still uncertain whether AI-ML algorithms have the capacity to accidentally demonstrate prejudice towards specific groups, such as women's communities, minority and marginalized populations, and other specialized classes. Fairness and transparency in artificial intelligence and machine learning applications are difficult to achieve. All individuals, including those who apply for loans and provide a valid email address, are subject to the possibility of being rejected. In the process of loan approval, artificial intelligence and machine learning models that are dependent on the decisions made by loan officers in the past may likewise exhibit bias. Therefore, it is of the utmost importance to supply reliable data training in order to nurture artificial intelligence and machine learning systems that are more resistant to the influence of being biased by humans.

The hybrid model, which combines AI-ML with other credit scoring methods or credit scoring methods that are based on AI-ML, is considered to be the most effective alternative alternative for evaluating credit scores, as stated in Table 5 of the comparison of credit scoring methodologies. When compared to other methods, logistic regression has a reduced impact. According to the findings of our investigation, the performance of the credit scoring model that is based on machine learning is greatly impacted by four primary characteristics.

In order to carry out the analysis, the following weightings were utilized: performance (0.3), robustness (0.2), accuracy (0.3), and data volume (0.2). When referring to the overall weight or the complete weight of anything, the phrase "one" is used appropriately.

Table.1 Denotes Bank performance grading scale values.

S/N	Rating	Range (%)	Description
1	rating 1	91-100	outstanding
2	rating 2	71-90	superior
3	rating 3	51-70	good
4	rating 4	31-50	average
5	rating 5	< 30	poor

A comprehensive and analytical literature analysis was done during business understanding to identify the bank's issues. After that, data mining fixed these issues. The purpose of this research was to examine how HR results affect company performance using surveys. The data was collected via a thorough questionnaire, in-person interviews, attentive observation, and three-year annual reports and account statements. Nigerian commercial banks employed "data mining" to recognize the enormous impact workers' abilities, attitude, and behaviour have on their work. This approach revealed hidden information and predicted. The bank's people resources should do well despite the competitive market, according to projections. Divide organizational performance into five levels—outstanding, superior, good, average, and poor—to turn the challenge into a data mining assignment. These levels were utilized to assess current staff data (Table 1). Supervised learning was used to create classification models from data with a known aim variable.



Fig.2 Denotes Formulate a lending strategy for a fictional banking client.

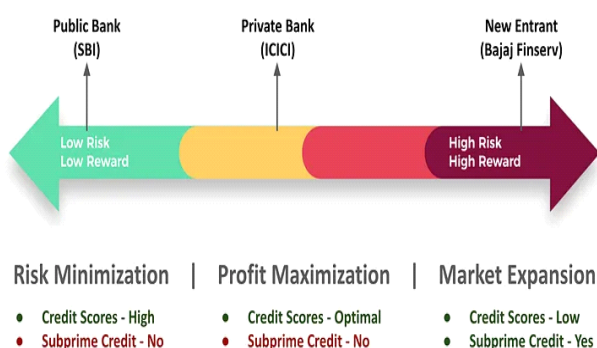


Fig.3 Denotes the scale of operations on banks.

The way banks make decisions is complicated by a lot of different factors as shown in fig.2 and 3. To begin, it's important to make it clear that having a longer credit past doesn't always mean you have a higher or better credit score for all borrowers. The second thing that needs to be thought about is the bank's size. People with good credit are not likely to use the services of a small lending

company. It makes things more difficult for this small lender to decide what to do.

V.CONCLUSIONS AND FUTURE DIRECTIONS

Recently, banks have prioritized machine learning for credit rating and risk assessment. These technologies can increase credit scoring accuracy and efficiency, which has major consequences for the sector. They may research large amounts of data to find patterns to improve projections. Many research have examined credit risk prediction benchmarking and evaluation methodologies. Banks use assessments to create policies and operations. However, unknown areas may be beneficial to research, underlining the need for continued research and growth in this sector. Banking also struggles to appropriately assess client creditworthiness. This assignment requires feature selection and machine learning to evaluate massive datasets with many variables. Building a credit-scoring prediction model using feature selection and machine learning requires finding the best method. For future growth, finding the most accurate feature selection method is crucial. Innovative ideas and technologies are needed to improve credit scoring systems' reliability and impartiality. Comprehensive data science platforms that can support bespoke artificial intelligence applications and models for many elements of a firm, including credit scoring AI, are another possibility. To conclude, machine learning in banking credit scoring must address non-performing loans and unreliable scoring systems. Equitable and realistic models require ongoing research and development.

REFERENCES

- [1] J. Zhang and Q. Sun, "Research on the financing cost of small and medium-sized enterprises by internet finance Open," *Open Journal of Social Sciences*, vol. 48, no. 25, pp. 95, 2017.
- [2] Y. Lin and C. Chen, "Research on enterprise financial risk evaluation based on association rules," *Friends Account*, vol. 96, no. 2365, 2017, pp. 32–35.
- [3] J. Liu, J.A. Xiuyi, et al., "Multi-label classification algorithm based on association rules mining," *Journal of Software*, vol. 28, no. 11, 2017, pp. 2865–2878.
- [4] G. R et al, "An Effective Construction of Multiple Viewpoint based Panorama Design using Digital Image Processing Scheme,"(ACCAI), Jan. 2022, Published, doi: 10.1109/accai53970.2022.9752626.
- [5] Sethy, P.K., Panigrahi, M., Vijayakumar, K. et al. Machine learning based classification of EEG signal for detection of epileptic seizure without snipping. *Int J Speech Technol* (2021). <https://doi.org/10.1007/s10772-021-09855-7>.
- [6] F. I, A. et al, "LoRaWAN Mesh Networks: An efficient mesh formation of LoRa devices in rural energy system," (ICESES), Chennai, India, 2022, pp. 1-8, doi: 10.1109/ICESES55317.2022.9914273.
- [7] S. Firdous and R. Farooqi, "Impact of internet banking service quality on customer satisfaction," *Journal of Internet Banking and Commerce*, vol. 121, no. 2201, 2017, pp. 1–17.
- [8] Venkatesan, K., Chandrasekar, A. & Ramesh, P.G.V. On-demand DWDM design using machine learning. *Soft Comput* 26, 6577–6589 (2022). <https://doi.org/10.1007/s00500-022-07181-x>

- [9] R. Nuthakki, P. Masanta and Y. T N, "Speech Enhancement based on Deep Convolutional Neural Network," 2021 Fifth International Conference on I-SMAC (IoT in Social, Mobile, Analytics and Cloud) (I-SMAC), 2021, pp. 1-6, doi: 10.1109/I-SMAC52330.2021.9640736.
- [10] H. A. AbdouGenetic, "Programming for credit scoring: The Case of Egyptian Public Sector Banks," Expert Systems with Applications, vol. 9, no. 36, pp. 11402-11417, 2009.
- [11] M. T, et al "IoT based Smart Robotic Design for Identifying Human Presence in Disaster Environments Using Intelligent Sensors," (AUTOCOM), Dehradun, India, 2024, pp. 399-403, doi: 10.1109/AUTOCOM60220.2024.10486106.
- [12] R. T. et al "Automated Crack and Damage Identification in Premises using Aerial Images based on Machine Learning Techniques," (I-SMAC), Dharan, Nepal, 2022, pp. 498-504, doi: 10.1109/I-SMAC55078.2022.9987391.